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Report

A Quantitative Genetic Signature of Senescence in a Short-Lived Perennial Plant

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Summary

The evolution of senescence (the physiological decline of organisms with age) poses an apparent paradox because it represents a failure of natural selection to increase the survival and reproductive performance of organisms. The paradox can be resolved if natural selection becomes less effective with age, because the death of postreproductive individuals should have diminished effects on Darwinian fitness [1, 2]. A substantial body of empirical work is consistent with this prediction for animals, which transmit their genes to progeny via an immortal germline. However, such evidence is still lacking in plants, which lack a germline and whose reproduction is diffuse and modular across the soma. Here, we provide experimental evidence for a genetic basis of senescence in the short-lived perennial plant *Silene latifolia*. Our pedigree-based analysis revealed a marked increase with age in the additive genetic variance of traits closely associated with fitness. This result thus extends to plants the quantitative genetic support for the evolutionary theory of senescence.

Results and Discussion

Senescence illustrates the ultimate failure of natural selection to optimize traits expressed in later life and is thought to come about because fewer and fewer individuals survive to contribute genes to the next generation as cohorts age [3]. Ultimately, any evolutionary explanation for senescence must presume the existence of genetic variation for aging in natural populations, i.e., that a decline with age in physiological condition linked to fitness traits has a genetic component expressed in terms of "gene-by-age" (G × A) interactions. The evolution of senescence along these lines is now widely accepted in animals. For instance, changes in longevity in response to artificial selection for earlier or later reproductive output have been induced in experimental populations of *Drosophila* [1], *Tribolium* [4], and *Caenorhabditis* [5]. Moreover, using a pedigree analysis, Wilson et al. [6] found evidence for G × A underlying the variation of life-history traits in both Soay sheep (*Ovis aries*) and red deer (*Cervus elaphus*), and Charmantier et al. [7] found equivalent G × A effects in swans (*Cygnus olor*). These and other studies clearly demonstrate the existence of a quantitative genetic basis for senescence in organisms with an immortal germline. However, in plants, which do not have a separated germline, but are instead characterized by modular reproduction that is diffuse across the soma, and where growth is often indeterminate, to our knowledge no study has yet examined the additive genetic basis of senescence.

Here, we tested for the presence of a G × A interaction underlying the expression of flower size and number in the dioecious plant *Silene latifolia*—two traits that are known to affect fitness [8], to be heritable and responsive to selection [9–11], and to vary with plant age [12]. The species grows preferentially in open habitats [13] and displays white flowers that are principally pollinated by night-flying moths. Although *S. latifolia* is a short-lived perennial, early measurement of floral traits is possible because plants usually start flowering during their first year [14] and reach a plateau in their flower production within about 5 months [12]. Our experiment involved observation of the phenotypes of individuals of *S. latifolia* as they aged in a glasshouse over the three generations of a crossing experiment; the members of each generation were randomly mated, and the established pedigree was recorded for analysis. We replicated our experiment three times, using genotypes from each of three widely separated natural populations in each replicate experiment. We also subjected half of all plants to a defoliation treatment at the commencement of flowering to allow comparison of stressed and unstressed plants. Data were analyzed using a pedigree-based random regression "animal model" (RRAM) [15–17] to test for the presence of additive genetic variation underlying aging rates in *S. latifolia*. Consistent with the earlier work cited above, we found that flower size and number declined significantly with age (Figure S1 available online).

A comparison of models that included or excluded G × A interactions for flower size and number in *S. latifolia* indicated an additive genetic component for rates of aging (Table 1). In particular, narrow-sense heritabilities for the intercept and the slope of the linear regression of calyx diameter on age were, respectively, 33% and 34%. For flower number, narrow-sense heritabilities for the intercept, the linear coefficient, and the quadratic coefficient defining the regression on age were, respectively, 32%, 46%, and 46%. Notwithstanding these global patterns, genetic variance components differed significantly among replicate populations, genders, and growing conditions in the glasshouse. Although these sources of variation were not the focus of our study, they were fully modeled and accounted for in all analyses.

A key difference between animals and plants is that plants do not have a germline and that their reproduction can be diffuse and indeterminate over an ever-extending soma, leading some to argue that they should not be subject to processes leading to the evolution of senescence [18, 19]. There is in fact substantial evidence that plants, even those reproducing clonally, do age [20–24], although discussions of the senescence of leaves [25, 26] are relevant only inasmuch as leaf senescence contributes to organismal death [27]. Plant physiology is known to be compromised by the same cellular processes that occur in animals, such as reactive-oxygen-based cell

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Table 1. Quantitative Genetics Random Regression Model Selection

	Deviance Information Criterion		
	Model 1	Model 2	Model 3
Calyx diameter	3,536	**3,460 (3,165)**	3,496
Number of flowers	7,372	7,370	**7,368 (7,367)**

Deviance information criterion (DIC) scores of quantitative genetics random regression models testing for gene-by-age (G × A) interactions for the average number of flowers produced per day and the average calyx diameter (mm) of flowers produced per day. The first model assumed no G × A by treating the genetic merit of individuals as a constant through age (model 1). The second model assumed a G × A interaction in the form of a linear additive genetic regression on age (model 2). The third model assumed a G × A interaction in the form of a degree two polynomial additive genetic reaction norm to age (model 3). For both traits, the model with the smallest DIC (in bold) provided the best fit to the data. For calyx diameter, treatment of the permanent environment and the residual components of variance as a linear function of age significantly improved the fit of model 2, which was significantly better than models 1 and 3. The DIC for this model is shown in parentheses. For the number of flowers, addition of a second-degree polynomial permanent environment and a residual component marginally improved the fit of model 3, which was better than models 1 and 2, albeit only marginally. See also Tables S1 and S2.

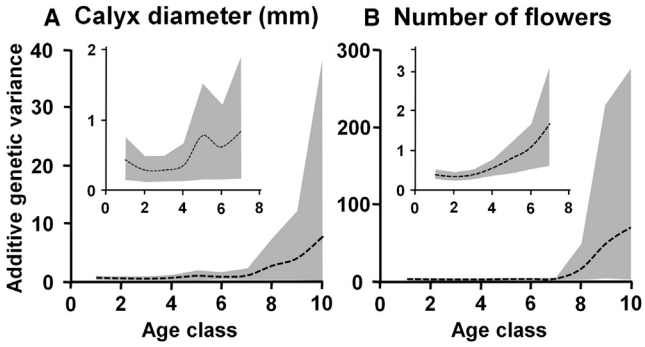


Figure 1. Age-Dependent Increase in Additive Genetic Variance of Mean Calyx Diameter and Number of Flowers

Raw estimates of additive genetic variance (V_A) were calculated for each age class, as generally done in character state approaches: these are represented by the dotted line for (A) the average calyx diameter of flowers produced per day and (B) the average number of flowers produced per day, with its 95% credible interval represented by the gray area (based on the density of parameter posterior distributions). The inset in each panel depicts the same data observed over the earlier ages of growth. See also Figure S1 for the age-dependent decline of the corresponding phenotypes with age and Table S1.

death [28] and problems of DNA replication that bring about telomere attrition [24, 29]. There is also some demographic evidence for senescence in plants [21, 30, 31], though patterns vary greatly among taxa, and some species even show negative demographic senescence [32, 33]. For example, Silvertown et al. [23] documented an increase in the mortality rates with age in about half of the 65 perennial plant species in their study. Similar age-specific mortality patterns have been found in experimental populations of the nonclonal perennial herb *Plantago lanceolata* [34–36], including higher mortality rates and reduced lifespan in plants under stress. Demographic evidence for senescence has also been found for number of perennial grasses and forbs [37]. Our finding that G × A interactions underlie the variation in fitness-related traits in *S. latifolia* now provides quantitative genetic evidence for the theory of the evolution of senescence in nonclonal plants.

As senescence evolves, two nonexclusive genetic mechanisms will leave their signature on the age-specific contribution of genetic loci underlying quantitative traits related to fitness. First, mutations reducing the physiological performance in old age can be propagated to the next generation because they are not expressed or are neutral early in life when the force of natural selection is strongest (the "mutation accumulation" hypothesis) [38]. Second, genes with deleterious effects in old individuals might nevertheless be selected because of their beneficial effects earlier in life (the "antagonistic pleiotropy" hypothesis) [18]. Under both hypotheses, the additive genetic variation underlying fitness traits is expected to increase in later life [2, 15, 39], but antagonistic pleiotropy can also result in a negative correlation between additive genetic variation expressed in early and later life [2, 15, 39]. Although such hypotheses have been elaborated and discussed largely with animals in mind, each of them is general and ought to apply to plants as well. Our findings are consistent with these fundamental ideas.

Our results also show that additive genetic variance increased with age for flower size and number (Figure 1). Here, the additive genetic variance underlying fitness traits was calculated separately for each age class using the character state approach (see the Supplemental Experimental Procedures for details). Both mutation accumulation and antagonistic pleiotropy may therefore be accountable for the evolution of senescence in *S. latifolia*. Additive genetic correlations between pairs of age classes, estimated using bivariate animal models, did not reveal negative correlations between early and late age classes, as might have been expected under antagonistic pleiotropy (Tables 2 and 3; see the Supplemental Experimental Procedures for details). Taken together, these results thus suggest a prevalent role of mutation accumulation in the evolution of senescence in *S. latifolia*, although antagonistic pleiotropy might also be involved in the observed increase in additive genetic variance and can therefore not be rejected as a potential mechanism for the evolution of senescence in *S. latifolia*. In the annual plant *Geranium carolinianum*, Roach [40] found phenotypic evidence for traits that increased fitness early in life and reduced fitness in older individuals, suggesting that antagonistic pleiotropy might play a role in senescence in this species.

Our observations were made on plants growing from germination to death over a period of approximately 16 months in each generation (including approximately 200 days of flowering); to what extent are they relevant to the evolution of a species that may survive for several years in the wild? As for any study of plants grown under nonfield conditions, the phenotypes we observed may have little direct bearing on trait evolution in the wild. In our experiment, plants were stressed toward early senescence by resource and space limitation within their pots, rendering an otherwise polycarpic plant effectively monocarpic. However, the crucial result of our study is not so much the phenotypic changes we observed among ages, but the underlying G × A interaction, because it most likely reflects a signature of the evolution of senescence over many generations in the field. The effect of resource depletion and the lack of space in small pots for root development probably had an effect on phenotypes that is confounded with age and should affect narrow sense heritability through its impact on the environmental component of phenotypic variation [41]. However, unless our experiment released the expression of otherwise cryptic genetic variation, which seems unlikely [42], we can think of no reason why

Table 2. Additive Genetic Correlations between Age-Specific Average Measures of Calyx Diameter

	Age Classes				
Calyx Diameter	1	2	3	4	5
2	0.590				
3	0.462	0.675			
4	0.461	0.447	0.268		
5	0.495	0.430	0.423	0.550	
6	0.104	0.356	0.106	0.177	0.759

All genetic correlations were estimated in bivariate animal models that included fixed effects of gender, population, and treatment and random permanent environmental and maternal effects.

Table 3. Additive Genetic Correlations between Age-Specific Average Numbers of Flowers Produced per Plant

	Age Classes				
Number of Flowers	1	2	3	4	5
2	−0.029				
3	0.001	−0.007			
4	−0.008	−0.025	0.026		
5	0.063	0.042	0.014	0.063	
6	0.087	−0.017	0.029	0.177	0.020

All genetic correlations were estimated in bivariate animal models that included fixed effects of sex, population, and treatment and random permanent environmental and maternal effects.

stressful experimental conditions should bias our analysis of the age-dependent additive *genetic* component of fitness traits. It would of course be interesting to know how additive genetic variance in fitness changes with plant age over a longer time span than that observed in our experiment.

The world's oldest living organism is claimed to be a clonal aspen whose seed may have germinated 10,000 years ago and that has persisted through the ongoing production and fragmentation of functionally independent ramets [43]. Such extreme examples of longevity, and the widespread incidence of clonal reproduction in plants, might suggest that some plants can ultimately avoid senescence—although an adoption of an entirely clonal habit will render them susceptible to the long-term genetic implications of the loss of sex and recombination [44] and to the accumulation of somatic mutations [20]. However, a decline in the efficacy of natural selection in older cohorts of any organism must necessarily lead to an age-related decline in reproductive performance. Our findings provide the first quantitative genetic evidence for this fundamental implication of the evolutionary theory of senescence in plants.

Experimental Procedures

Full details of our experimental setup, methods, sample sizes, and analytic approach are given in the online Supplemental Experimental Procedures for this article and are summarized briefly here.

Seed was sampled from open-pollinated fruits in three natural populations: Salzburg, Austria; Potsdam, Germany; and Oxford, United Kingdom. Plants were grown from seeds over three generations in a common garden experiment in a glasshouse at the University of Oxford (n = 133 for Salzburg, n = 72 for Oxford, and n = 78 for Potsdam). Within populations and generations, individuals were crossed at random, and plant identities were recorded for inclusion in a pedigree over the course of three generations, with n = 203, 47, and 33 plants measured in the first, second, and third generations, respectively. During the flowering period, each new open flower was counted and tagged, and its calyx diameter was measured. Our analyses are based on measurements of 20,988 flowers on a total of 283 plants over the course of their lives. The youngest plant to produce flowers was 30 days old, whereas the oldest plant measured was 590 days old. In each generation, plants characterized by different ages flowered for approximately 200 days. The range of plant ages was divided into ten age classes. For each plant in each age class, we calculated the average number of flowers produced and the average calyx diameter of flowers.

To test for gene by age interactions (G × A), we conducted a RRAM. Phenotypic variance (V_P) was decomposed as $V_P = V_A + V_M + V_{PE} + V_R$, where V_A is the additive genetic variance, V_M the maternal variance, V_{PE} the permanent environment variance, and V_R the residual variance.

This animal-model approach was included within the random regression framework, where the additive genetic merit of individuals, $a_i(Age)$, was modeled as a function of age. The RRAM model was specified as $p_{t,i} = m + pop + sex + env + AGE + a_i(Age) + pe_i + m_k + e_i$, where $p_{t,i}$ is the phenotype of individual i at time t, m is the phenotypic average, pop is a fixed population effect (populations A, E, or G), sex is a fixed sex effect (male or female), env is a fixed environmental effect (stress by defoliation versus no stress), and AGE is a fixed effect of age included as a factorial factor (age t).

We built three models that differed in the age-dependent function of the genetic merit, with $a_i(Age)$ for each trait specified as a polynomial function of age with degree x, where x = 0, 1, and 2 for the three models, respectively. Model selection was based on the DIC score of each model. In order to establish age-related patterns in additive genetic variation, we used a series of univariate quantitative genetics animal models to estimate additive genetic variance per each age class. After reducing the number of age classes to six, we used bivariate animal models to estimate additive genetic correlations between pairs of age classes and to build the corresponding correlation matrix. Annotated R code protocols to run the random regression models in R are available from the corresponding author upon request.

Supplemental Information

Supplemental Information includes Supplemental Experimental Procedures, one figure, and three tables and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2014.02.012.

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